

## Low-Cost Polymeric Optical Fiber Sensor for Physiological Measurement

**María Gaitán-Padilla<sup>1</sup>, Maria J. Pontes<sup>1</sup>, Maxwell E. Monteiro<sup>2</sup>, Carlos A. Cifuentes<sup>3, 4</sup>,  
Camilo A. R. Diaz<sup>1</sup>**

<sup>1</sup> Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil, maria.padilla@edu.ufes.br, maria.pontes, camilo.diaz[@ufes.br]

<sup>2</sup> Instituto Federal do Espírito Santo (IFES), Campus Serra, Serra, Brazil, maxwell.monteiro@ifes.br

<sup>3</sup> Bristol Robotics Laboratory, University of the West of England, Bristol, UK,  
carlos.cifuentes@uwe.ac.uk

<sup>4</sup> The School of Engineering, Science and Technology, Universidad del Rosario, Bogotá D.C.,  
Colombia.

**Abstract:** Physiological monitoring is important in different scenarios for improving the prevention and early detection of heart and respiratory diseases. This study evaluated a low-cost fiber-optic physiological sensor developed for stress measurement. This sensor was developed to avoid invasive sensors that require physical contact with the skin for measurement, thereby generating discomfort. Five healthy subjects participated in a validation test that compared the proposed sensor with a known commercial sensor. The pulse and respiration detection were performed correctly using the developed sensor. Moreover, mean errors of 8.34% and 9.97% were obtained for pulse and respiratory rates, respectively, when compared with the reference values. Therefore, it is necessary to improve the pulse and breathing errors with the proposed low-cost optical fiber sensor for its implementation in future potential studies of respiratory and heart rate detection.

**Keywords** Heart rate, Optical fiber sensor, Physiological measurement, Respiratory rate.

### 1. Introduction

Physiological measurements are used to determine the state of health with respect to the vital functions of the human body. Doctors perform this measurement in a clinical setting during auscultation, oximetry, and blood pressure measurement [1]. However, more rigorous control and monitoring are recommended in individuals with cardiac and respiratory diseases to prevent extreme conditions that can considerably affect health, such as cardiopathies, and asthma episodes [2]. This monitoring should be performed at home for all activities of daily living, in real life, and in high-risk scenarios.

In addition, monitoring has recently been promoted for early diagnosis and prevention of diseases [2]. Sensors such as those focused on sports environments have also been developed for measurement and monitoring, where the influence of factors such as movement, sweat, and humidity are considered [3]. These sensors are often attached to other devices, such as the Garmin smartwatch and the Apple Watch, with sensors attached to watches with an additional band in the chest under clothes [4]. However, by including this function and measuring bands, the prices of these devices increase and become a limitation.

Clinical sensors such as sphygmomanometers [5], and commercial sensors such as Shimmer [6] have been developed for physiological measurements. Recently, fiber optic-based sensors have been used for physiological measurement and constant monitoring because of their sensitivity, small size, flexibility, potential use in wearable sensors, continuous monitoring systems, and electromagnetic

immunity [5]. However, fiber Bragg grating optical sensors, which are the most commonly used, require interrogation systems that are costly [7], generating a limitation in this sensing method. On the other hand, polymer optical fiber (POF) sensors based on light intensity detection have been an alternative approach for wearable sensors, which are easy to fabricate, flexible, and require simple and low-cost interrogation systems. The use of these sensors in the detection of physiological variables in wearable sensors has mainly been designed for measuring heart, and respiratory rates [8], among other health applications [9].

In addition, light-based sensors have been developed with the concept of photoplethysmography, and have been evaluated in different body locations in measurement of multiple physiological variables, which is a benefit of computational methods to differentiate biosignals based on their characteristic frequency [10]. Within these studies, detection errors are estimated when compared with clinical or commercial reference sensors, for which the errors vary from 1.5 % to more than 50 % [10].

Moreover, studies on the physiological response to stress states have been conducted [5]. In addition, among the consequences of not controlling stress is the generation of psychological illnesses such as anxiety and depression. Likewise, cardiac and respiratory diseases can be diagnosed and monitored, for which a robust, low-cost, and wearable sensor is needed.

The present study aimed to compare the proposed polymer optical fiber sensor with a commercial wearable sensor in physiological monitoring. The main objective was to validate the physiological measurement of the proposed low-cost wearable physiological sensor for implementation in future potential applications.

## 2. Methodology

This study aims to validate a sensor developed for wearable physiological measurements using low-cost fiber-optic technology. Initially, a sensor was developed with a polymeric optical fiber that had two sensitive areas that allowed variations in the light power measured with a photodetector for the simultaneous measurement of two physiological signals. Based on this development, a detailed explanation of the sensor's operation, and respiratory and heart rate measurements are provided. Subsequently, an experimental protocol compared two physiological measurement sensors: a commercial sensor used in continuous monitoring in real scenarios, and the proposed optical sensor.

### 2.1 The developed sensor

Initially, the proposed optical fiber sensor for physiological variables consisted of four main subsystems, as shown in Figure 1a: the POF-based sensor, electronics, the Digital Acquisition System (DAQ), and a computer for processing and obtaining the signal. The first subsystem is the sensing part, detailed in Figure 1b, which consists of a POF, polymeric elastic filament, photodetector, and light source. The two sensitive areas proposed were equally manufactured and spaced 3.5 cm apart for placement on the chest, as shown in Figure 1a. The working principle for the physiological measurement starts when the light from the light emitting diode (LED) enters to the POF, and the sensitive zone is made of a polymeric filament with a silicon cover and coupled to the POF with a 3D printed piece, generating a light loss variation due to the flexibility of the polymeric material. This light loss varies depending on the expansion and contraction provoked by the pulse and breathing movements in the chest, as measured by the photodetector at the end of the POF. The voltage variation was obtained in the photodetector output, which was measured using an electronic circuit to maximize the variation. The electronics were then connected to a DAQ, which converts the analog voltage variation into a digital signal that can be read in the computer with the MATLAB program as the processing tool

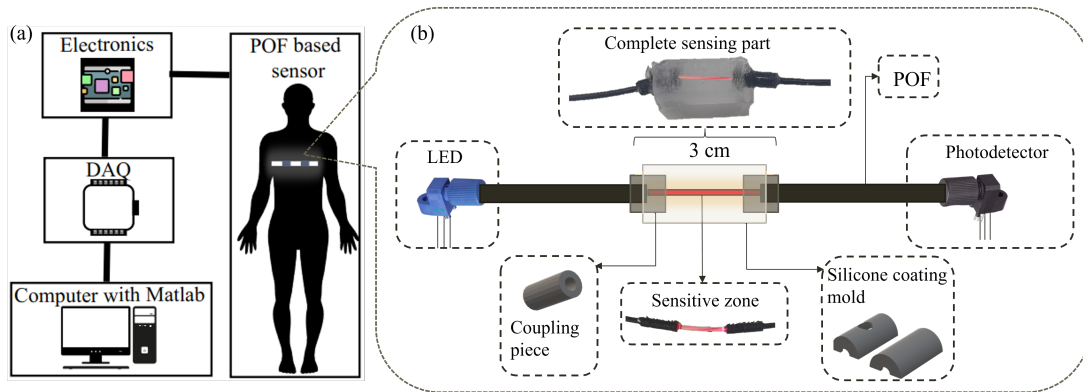


Figure 1. Working principle schematic of the proposed low-cost optical fiber sensor for breathing and pulse rate monitoring. (a) The four subsystems: are the POF sensor, the electronic circuit, the DAQ, and the preprocessing tool with Matlab installed. (b) The working principle diagram of the POF sensor subsystem explains the components included in the sensing part.

for obtaining and analyzing the signals. The complete system is illustrated in Figure 1. The sensor was added to a Velcro to make it adaptable to any participant. The sensor was tested to identify the voltage variations generated by the signal targets to be measured. Therefore, the two sensing parts had the best performance in the variation and measurement of physiological signals.

## 2.2 Pulse and respiratory signal

MATLAB was used to process the signals in the fourth subsystem. The pulse signal was first filtered using a Butterworth third-order bandpass filter (1-3.5 Hz) [8], and the respiratory signal was first filtered using a Butterworth third-order bandpass filter (0.1-0.8 Hz) [8]. The `filtfilt` command avoids the offset provoked by the filter in both signals. In addition, to obtain the heart and respiratory rates, the `Findpeaks` function of MATLAB was used, using a minimum peak distance of 60 to avoid consecutive incorrect peaks from noise still present in the filtered signal and filter influence.

## 2.3 Validation protocol

To validate the proposed sensor, a protocol was developed to compare the heart rate with that of a commercial sensor used in real-world scenarios of physiological monitoring. The protocol was approved by the Lions Club School's ethics committee as the sensor validation first step, all participants were informed of the nature of the study and signed the consent form.

A commercial Garmin Forerunner 920XT sensor was used, and a cell phone was needed to extract the information recorded with the watch and wearable band. Breathing was manually counted from the videos for the respiratory rate reference value. The proposed sensor was adapted to a flexible Velcro to be placed and fitted to each participant's chest at axillary height and tightened to the participant's liking and comfort. The sensor was validated with a one-minute test in five subjects (one woman and four men) to obtain and compare the signals to the commercially available sensor. Initially, the sensors were located, the commercial sensor was located under the clothing and the proposed sensor over the clothing. Then, a brief test was performed with both sensors to ensure operation during the one-minute test. Finally, the test was recorded to obtain the respiratory reference rate. Some questions regarding comfort and intention of use were asked as brief feedback for future improvements, with an adapted questionnaire based on the Unified Theory of Acceptance and Use of Technology (UTAUT) using a Likert scale from 1 to 5. Errors in the measured heart and respiratory

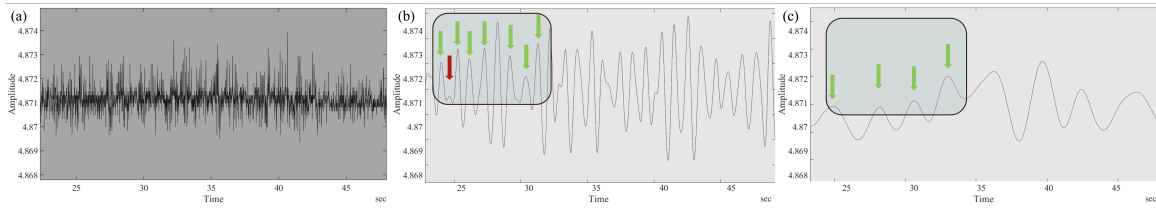


Figure 2. Preprocessing to simultaneously obtain the respiratory and pulse signals for extracting the peaks used for the heart and respiratory rates. (a) The raw signal is obtained from the sensor. (b) Filtered signal to obtain the pulse signal with some correct green and incorrect red peaks. (c) Filtered signal to obtain the respiratory signal with some correct peaks marked in green.

rates were obtained with respect to the commercial reference sensor. Using this value and graphical analysis, the signals obtained with the sensor were evaluated.

### 3. Results and discussion

A sensor to measure the pulse and respiratory signals was developed to monitor simultaneously and accurately with low-cost polymeric optical fiber. These types of sensors have been previously developed and evaluated in dynamic scenarios [8], and with FBG optical sensors compared to reference sensors [11]. Using the proposed methodology to validate the low-cost wearable optical fiber sensor, five one-minute signals were analyzed from the subjects. An example of the raw, pulse, and respiratory signals obtained from the wearable system is shown in Figure 2, where some correctly detected peaks are marked with green arrows, and incorrectly detected peaks are marked in red. The respiratory signal in the example had only correctly detected peaks.

Both sensors' measurements were taken synchronously for one minute. The results for the five subjects are presented in Table 1. It is important to mention that the comparison proposed is with wearable commercial sensors, so the commercial sensor had an error percentage relative to the real value. For the respiratory reference, the video used for the reference estimation also had an error in the real respiratory rate.

| Subject | Pulse (bpm) |           |         | Breath (breath pm) |           |         |
|---------|-------------|-----------|---------|--------------------|-----------|---------|
|         | Sensor      | Reference | Error % | Sensor             | Reference | Error % |
| 1       | 62          | 73        | 15.07   | 16                 | 15        | 6.67    |
| 2       | 57          | 63        | 9.52    | 18                 | 17        | 5.88    |
| 3       | 60          | 57        | 5.26    | 16                 | 14        | 14.29   |
| 4       | 66          | 62        | 6.45    | 18                 | 16        | 12.50   |
| 5       | 70          | 74        | 5.40    | 21                 | 19        | 10.53   |

Table 1. Validation of the low-cost fiber optic sensor against the reference value and the heart and breathing rate detection error.

Considering the graphical behavior of the pulse signal after the filtering process shown in Figure 2b, errors can be seen in the automatic peak detection, where the peaks of the pulse signal correspond to the R peak of the heart cycle and the incorrect peaks detected are due to noise and the filter influence. The pulse and breathing detection with the proposed fiber optic sensor had a mean error in the pulse of 8.34 % and in the breathing of 9.97 %. This is acceptable considering other physiological detection studies with fiber optic sensors that report errors of 5.5 bpm compared with different commercial physiological detection devices [11]. In addition, there have been comparisons with clinical

devices for pulse rate estimation obtained from a clearer ECG signal [5], which is a comparison with the real pulse rate value. However, there was a clear difference in the inter-subject detection, comparing the results of the pulse error between Subjects 1 and 3, with a difference of 9.41%. In terms of respiratory error, comparing Subject 3 with Subject 2, there is a difference of 8.41%. Studies have evaluated the inter-subject pulse detection error in specific applications, such as stress classification, which is important for the analysis of the physiological response of a specific subject [12].

Considering the error of the commercial sensor used for this measurement, where in some cases, the placement of the commercial sensor generated difficulties by not generating measurements in some participants, a clinic sensor would be a better option for future studies as a reference. Likewise, it is necessary to improve the sensor's sensitivity to generate a clearer pulse and breathing wave and the processing methods for peak detection. This will facilitate obtaining features such as heart rate variability (HRV) extracted from the pulse signal for future biosignals-related disease detection and monitoring.

Finally, brief feedback was obtained after the study from each participant with a short questionnaire based on UTAUT. The main comments highlighted the comfort of the proposed sensor, the difference between both sensors and the intention to use it in real-life situations adapted in t-shirts. This has been achieved previously with other experimental sensors added to fabric fibers [13]. Moreover, the questionnaire results are not presented because the number of participants was not significant to determine if the sensor was comfortable or if it is an intention of use, but it gave brief feedback for the design.

#### 4. Conclusions

The developed sensor is low-cost, wearable, and based on optical fiber technology, where the main source for working is light. The sensor can be adjusted for any participant and is comfortable based on the feedback of all subjects. In the validation of the sensor, an average error of 8.34% was obtained in the pulse measurement, and an average error of 9.97% was obtained in the respiratory measurement, indicating the need to improve the proposed sensor to increase its sensitivity and minimize this error in the validation. However, this is a considerable error for heart rate and respiratory rate detection compared to other commercial wearable sensors that are costly and uncomfortable when used with skin contact.

For future work, the sensor needs to be improved for implementation in stress and anxiety classification studies, or for identifying respiratory or cardiac diseases. The signal processing can be better by decreasing the incorrect peak detection that affects the heart rate estimation, this process will be made with AI methods found in the literature.

#### 5. Acknowledgments

This work is partially supported by FAPES (209/2018 - Edital Especial CPID). Camilo Diaz acknowledges the financial support of FAPES (459/2021), CNPq (310668/2021-2), and MCTI/FNDCT/FINEP (2784/20).

#### References

- [1] Jiajia Song, Xianping Huang, Shenzhi Zhao, Jiajia Chen, Ruiheng Chen, Guowei Wu, and Zhangye Xu. Diagnostic value of pulse oximetry combined with cardiac auscultation in screening congenital heart disease in neonates. *Journal of International Medical Research*, 49(5):03000605211016137, 2021.



- [2] Brendan McCluskey, Michael Addis, Bartholomew J Tortella, and Robert F Lavery. Out-of-hospital use of a pulse oximeter to determine systolic blood pressures. *Prehospital and Disaster Medicine*, 11(2):105–107, 1996.
- [3] Dhruv R Seshadri, Ryan T Li, James E Voos, James R Rowbottom, Celeste M Alfes, Christian A Zorman, and Colin K Drummond. Wearable sensors for monitoring the physiological and biochemical profile of the athlete. *NPJ digital medicine*, 2(1):72, 2019.
- [4] David Hernando, Surya Roca, Jorge Sancho, Álvaro Alesanco, and Raquel Bailón. Validation of the apple watch for heart rate variability measurements during relax and mental stress in healthy subjects. *Sensors*, 18(8):2619, 2018.
- [5] Shouhei Koyama, Hiroaki Ishizawa, Satoshi Hosoya, Takashi Kawamura, and Shun Chino. Stress loading detection method using the fbg sensor for smart textile. *Journal of Fiber Science and Technology*, 73(11):276–283, 2017.
- [6] Jenny Nguyen, Robyn E Cardy, Evdokia Anagnostou, Jessica Brian, and Azadeh Kushki. Examining the effect of a wearable, anxiety detection technology on improving the awareness of anxiety signs in autism spectrum disorder: a pilot randomized controlled trial. *Molecular autism*, 12(1):1–8, 2021.
- [7] Christian Broadway, Rui Min, Arnaldo Gomes Leal-Junior, Carlos Marques, and Christophe Caucheteur. Toward commercial polymer fiber bragg grating sensors: Review and applications. *Journal of Lightwave Technology*, 37(11):2605–2615, 2019.
- [8] Arnaldo G Leal-Junior, Camilo R Díaz, Cátia Leitão, Maria José Pontes, Carlos Marques, and Anselmo Frizera. Polymer optical fiber-based sensor for simultaneous measurement of breath and heart rate under dynamic movements. *Optics & Laser Technology*, 109:429–436, 2019.
- [9] Arnaldo G Leal-Junior, Camilo AR Diaz, Letícia M Avellar, Maria José Pontes, Carlos Marques, and Anselmo Frizera. Polymer optical fiber sensors in healthcare applications: A comprehensive review. *Sensors*, 19(14):3156, 2019.
- [10] Sally K Longmore, Gough Y Lui, Ganesh Naik, Paul P Breen, Bin Jalaludin, and Gaetano D Gargiulo. A comparison of reflective photoplethysmography for detection of heart rate, blood oxygen saturation, and respiration rate at various anatomical locations. *Sensors*, 19(8):1874, 2019.
- [11] Cátia Tavares, Cátia Leitão, Daniela Lo Presti, MF Domingues, Nélia Alberto, Hugo Silva, and Paulo Antunes. Respiratory and heart rate monitoring using an fbg 3d-printed wearable system. *Biomedical Optics Express*, 13(4):2299–2311, 2022.
- [12] Roger Selzler, Adrian DC Chan, and James R Green. Impact of subject-specific training data in anxiety level classification from physiologic data. In *2021 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, pages 1–6. IEEE, 2021.
- [13] Lie Wang, Liyuan Wang, Ye Zhang, Jian Pan, Shangyu Li, Xuemei Sun, Bo Zhang, and Huisheng Peng. Weaving sensing fibers into electrochemical fabric for real-time health monitoring. *Advanced Functional Materials*, 28(42):1804456, 2018.